

Generation Interconnection System Impact Study Report

For Queue Project Q20-1506

Imhoff Tap 1 138 kV

June 22, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1506. The Interconnection Customer proposes a 55.2 MW battery energy storage facility interconnecting at the Imhoff Tap 1 138 kV substation in the Virginia area. The request was received on June 28, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 557 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,072,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
NEELY – VC PORTERDALE HYDROELECTRIC 138 KV line	MICHIGAN POWER LP – SKIFFES CRK 138 KV	131.2	277	7.1
BLUE RIDGE – NEELY 138 KV	MICHIGAN POWER LP SUBSTATION – SKIFFES CRK 138 KV line	128.9	295	7.1
IMHOFF TAP 1 – FUDGE HOLLOW 138 KV	BATH COUNTY – LONGVIEW2 345 KV	122.6	211	79.8
IMHOFF TAP 1 – OLA 138 KV	BATH COUNTY – LONGVIEW2 345 KV	115.4	181	20.2
OAK CRK STA – WADSWORTH 138 KV	DAYTON – CHATOM 345 KV Line	108.7	328	5.4
OAK CRK STATION SUBSTATION – SANS SOUCI 138 KV	DAYTON – CHATOM SUBSTATION 345 KV L	107.0	372	5.4
HARRISONBURG – SANS SOUCI 138 KV	DAYTON – CHATOM SUBSTATION 345 KV	107.0	372	5.4
LEAVITT – CHANCELLOR ROW 345 KV	LEAVITT – BIRCHWOOD SUBSTATION 345 KV	101.3	2,057	5.6
S BELLEVILLE SUBSTATION – NICHOLS 138 KV	PAGE – PURDY 138 KV line	101.1	506	5.1
ST LOUIS – NICHOLS 138 KV	PAGE – PURDY 138 KV L	101.0	512	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NEELY – VC PORTERDALE HYDROELECTRIC 138 KV	Rebuild line with higher-capacity conductor	\$2,504,000	2% (\$38,000)
BLUE RIDGE – NEELY 138 KV	Rebuild line with higher-capacity conductor	\$3,644,000	2% (\$55,000)
IMHOFF TAP 1 – FUDGE HOLLOW 138 KV	Rebuild line with higher-capacity conductor	\$5,060,000	49% (\$2,458,000)
IMHOFF TAP 1 – OLA 138 KV	Rebuild line with higher-capacity conductor	\$7,180,000	19% (\$1,384,000)
OAK CRK STA – WADSWORTH 138 KV	Reconductor line	\$2,421,000	1% (\$27,000)
OAK CRK STATION – SANS SOUI 138 KV line	Reconductor line	\$1,139,000	1% (\$13,000)
HARRISONBURG – SANS SOUCI 138 KV	Reconductor line	\$890,000	1% (\$10,000)
LEAVITT – CHANCELLOR ROW 345 KV	Reconductor line	\$10,053,000	0% (\$33,000)
S BELLEVILLE SUBSTATION – NICHOLS 138 KV	Reconductor line	\$1,437,000	2% (\$22,000)
NICHOLS – ST LOUIS 138 KV Line	Reconductor line	\$2,043,000	2% (\$32,000)
	Total allocated to project		\$4,072,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.